

TTMPC INTEGRATED MANAGEMENT & FINANCE SYSTEM END-TO-END PROCESS FLOW

Tubungan Teachers' Multi-Purpose Cooperative (TTMPC)
Integrated Management and Finance System with Parallel Risk Prediction,
Debt Capacity Framework, and Loan Demand Forecasting

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PHASE 1: ENTRY & AUTHENTICATION

Non-Members / First-Time Visitors

[1] Website Landing Page

[2] Role Selection (Member vs Non-Member)

NON-MEMBER PATH:

- View Loan Services (Consolidated, Bonus, Emergency, KOICA)
- View Savings Services (Regular, Special Programs)
- View Withdrawal/Deposit options

MEMBER PATH:

- Member Login (existing member)
- Sign Up (new applicant)
- Or continue as Non-Member

New Applicant Flow

[1] Sign Up Page

[2] Membership Form Submission

- Personal Data Sheet (bio, contact, occupation)
- Income Information
- Supporting Documents Upload

[3] Backend Processing

- Application recorded in Supabase
- Membership ID generated
- Status: PENDING_REVIEW

[4] BOD Approval Process (handled by Board Officers)

Existing Member Login

[1] Login Page (Email/Password)

[2] Authentication via Supabase

- Account lookup in member_account table
- Session creation
- Role assignment: Member, Bookkeeper, BOD, Manager, Treasurer, Cashier

[3] Redirect to Role-Based Dashboard

PHASE 2: MEMBER PORTAL

Member Dashboard - Main Hub

- Personal Profile Summary
- Account Balance Overview
- Active Loans & Schedules
- Savings Accounts Status
- Recent Transactions

A. LOAN SERVICES

[1] Member Loan Portal

[2] View Available Loan Types:

- Consolidated Loan (merges existing loans)
- Bonus Loan (performance-based)
- Emergency Loan (quick approval)
- KOICA Loan (special program)

[3] System Auto-Fill Pre-fill Data:

- Member personal info, Income records, Employment details

[4] Loan Computation Engine:

- Calculate Processing Fee
- Generate Repayment Schedule
- Display Gross vs Net Loan Amount
- Show Monthly Amortization

[5] Member Reviews & Submits with Digital Signature

[6] Confirmation Page & PDF Download

B. SAVINGS SERVICES

- Regular Savings Account (View Balance, Deposit, Withdraw)
- Special Savings Program (Manage Targeted Savings)
- Statement of Account (View historical transactions)

PHASE 3: ADMINISTRATIVE WORKFLOWS

BOD (Board of Directors) Portal

Member Approval Management:

- Review pending applications
- Verify personal data
- Approve/Reject membership

Training Attendance Tracking:

- Record member attendance
- Approve completion status
- Determine membership eligibility

BOOKKEEPER Portal

Member Records Management:

- Full member database, Search & filter capabilities

Loan Application Processing:

- Review incoming applications
- Pre-assess loan feasibility
- Submit to MIGS for scoring

MIGS (Automated Scoring System):

- Multi-criteria evaluation (Score 0-100)
- Risk prediction using Logistic Regression

Loan Approval Processing:

- Review scores and risk predictions
- Forward to Manager for final decision

Payment Records & Accounting:

- Record payments, GL entries, Account reconciliation

MANAGER Portal

- Loan Approval Authority (Final decision making)
- Member Management (Status updates, Co-maker relationships)

TREASURER Portal

- Loan Disbursement Processing (Check/transfer payments)
- Treasurer Accounting (Record transactions, Cash management)
- Final Approval Queue (Verification before disbursal)

PHASE 4: BACKEND PROCESSING & INTELLIGENCE

Supabase Database Layer

Core Tables:

member_account, member_applications, personal_data_sheet
member_profile, member_savings_accounts
loan_applications, loan_payments, loan_schedules
attendance_logs, grocery_transactions

Intelligent Decision Support Features

MIGS Scoring Engine:

- Multi-Criteria Evaluation (Personal, Income, Debt, Payment History)
- Output: Score 0-100 with risk classification

Parallel Risk Prediction:

- Logistic Regression Model
- Probability of default calculation

Debt Capacity Framework:

- Calculate member's ability to repay
- Assess net take-home pay and total debt
- Repayment Stress Index visualization

Loan Demand Forecasting:

- Time-Series Analysis for seasonal patterns

PHASE 5: LOAN LIFECYCLE (DETAILED)

Complete Loan Journey

- [1] LOAN APPLICATION: Member submits form with auto-filled data
- [2] BOOKKEEPER REVIEW: Review documents, Verify info, Submit to MIGS
- [3] AUTOMATED MIGS SCORING: Evaluation, Score 0-100, Risk level assigned
- [4] MANAGER APPROVAL: Final decision based on scores and risk prediction
- [5] TREASURER VERIFICATION: Final check, Fund availability, Authorize disbursal
- [6] CASHIER DISBURSAL: Release funds to member, Record transaction
- [7] LOAN MANAGEMENT: Track amortization, Monitor payment schedules
- [8] PAYMENT COLLECTION: Cashier records payments, Update balances
- [9] LOAN COMPLETION: Balance = 0, Mark as paid in full, Archive

PHASE 6: SAVINGS & CBU LIFECYCLE

Savings Management Flow

- [1] Member Initiates Deposit/Withdrawal
- [2] Goes to Cashier (at cooperative office)

DEPOSIT Process:

- Verify member ID
- Accept cash/check
- Record in system, Update balance
- Issue receipt

WITHDRAWAL Process:

- Verify ID, Check balance
- Record withdrawal, Update balance
- Disburse cash, Issue receipt

Cooperative Business Unit (Grocery) Operations

- [1] Member buys at CBU (Grocery)
- [2] Transaction recorded via POS system
- [3] Reconciliation (Cashier/Bookkeeper):
 - POS data synced to main system
 - Grocery ledger updated
 - Cross-check transactions
- [4] Financial Impact:
 - Grocery credit tracked in member account
 - Offset against savings/loan
 - Used in debt capacity calculations

New Member Membership Lifecycle

- [1] APPLICATION: Fill membership form with personal data
- [2] BOD REVIEW & TRAINING: Verify application, Require 1st Training
- [3] MEMBERSHIP APPROVAL: BOD votes, If training attended: Approve
- [4] ACTIVE MEMBER: ID assigned, Account activated, Services available

PHASE 7: REPORTING & ANALYTICS

Available Reports

Bookkeeper Reports:

- Member Records, Loan Applications, MIGS Scores
- Payment Collections, Savings Transactions
- Accounting Reports, Audit Trail

Manager/Treasurer Reports:

- Loan Portfolio, Default Risk Summary
- Disbursement Report, Collection Status
- Financial Position Report

Strategic Analytics (Decision Support):

- Loan Demand Forecast, Member Risk Dashboard
- Debt Stress Index, Default Probability Ranking

KEY DECISION POINTS IN THE SYSTEM

1. NEW APPLICANT?

YES: Start Membership Application > BOD Approval > Account Created

NO: Login > Go to Role Dashboard

2. MEMBER WANTS LOAN?

Pre-assessment > MIGS Scoring > Risk Analysis

If Risk HIGH: Reject or require co-maker

If Risk ACCEPTABLE: Manager Approval > Disbursement

3. LOAN PAYMENT DUE?

On time: Record payment, reduce balance

Late: Flag delinquency, send reminder

Default: Escalate, trigger collection procedures

4. SAVING WITHDRAWAL REQUEST?

Balance sufficient: Process withdrawal

Insufficient funds: Deny or inform member

5. GROCERY TRANSACTION?

Member has credit: Deduct from account

No credit: Reject transaction

Reconcile in main system regularly

SYSTEM ARCHITECTURE & DATA FLOW

Technology Stack

Frontend: React + Vite (Single Page Application)

Backend: FastAPI (Python-based REST API)

Database: Supabase PostgreSQL (Real-time, Cloud-hosted)

Authentication: Supabase Auth (Email/Password)

External Services: Resend API (Email), POS System (Grocery)

Data Flow Architecture

[1] Frontend Components (React)

[2] API calls via Supabase Client

[3] FastAPI Backend (main.py)

- Business Logic Processing
- Risk calculations, PDF generation

[4] Supabase PostgreSQL Database

- Persistent storage, Data relationships

[5] External Services:

- Resend API (Email notifications)
- POS System (Grocery transactions)

Role-Based Access Control

Member: View own loans/savings, Submit applications, Payment tracking

Bookkeeper: Process applications, MIGS scoring, Manage records, Reports

BOD: Approve memberships, Track training, Manage lifecycle

Manager: Loan approval authority, Final decisions, Member management

Treasurer: Loan disbursement, Financial control, Payment processing

Cashier: Savings deposits/withdrawals, Loan payments, POS, Daily cash

INTELLIGENT SYSTEM CAPABILITIES

1. Parallel Risk Prediction
Logistic Regression to estimate default probability
2. Payment Capability Assessment
Evaluates ability to repay based on net income
3. MIGS Scoring Engine
Multi-criteria evaluation (0-100 scale)
4. Loan Demand Forecasting
Time-Series Analysis for seasonal patterns
5. Repayment Stress Index
Visual tool for monitoring financial burden
6. Multi-Loan Tracking
Concurrent loans with aggregate debt monitoring
7. Automated Decision Support
Reduces bias, speeds approval, ensures consistency

SYSTEM SUMMARY

The TTMPIC Integrated Management and Finance System is a comprehensive Decision Support System (DSS) that transforms manual cooperative operations into a centralized, intelligent, and predictive platform.

CORE FEATURES:

- Complete member lifecycle management from application to active status
- Multi-type loan management (Consolidated, Bonus, Emergency, KOICA)
- Intelligent risk assessment and default prediction
- Automated MIGS scoring engine (0-100 scale)
- Debt capacity and repayment stress analysis
- Savings account management and transaction tracking
- Grocery POS integration and reconciliation
- Role-based access control (6 user roles)
- Comprehensive reporting and analytics
- Real-time decision support for loan approvals

PROCESS FLOW:

All transactions flow through: Entry > Authentication > Role Portal > Transaction Processing > Backend Intelligence > Database Storage > Reports > Member Notifications

KEY DIFFERENTIATORS:

This system combines operational management, risk prediction, financial analysis, and strategic forecasting into a single, cohesive decision support system designed specifically for cooperative financial management.